

PROJECT REPORT

Task for the Course: IWNF02_E – Project: Agile Software Engineering

CONTENT

1. Task.....	2
1.1 Task 1: Customer Self-Service Portal_SwissAXA.....	2
1.2 Task 2: Campaign Management Software_Bayer-Wesphalian	3
1.3 Task 3: Learning Management System_NordWest Business School	4
2. Additional information for the evaluation of the Project Report.....	6
3. Tutorial Support	6
References.....	6

1. TASK

You can choose from the following tasks for your project report. Please choose one of them to work on in your project report.

Note on copyright and plagiarism:

Please take note that IU Internationale Hochschule GmbH holds the copyright to the examination tasks. We expressly object to the publication of tasks on third-party platforms. In the event of a violation, IU Internationale Hochschule is entitled to injunctive relief. We would like to point out that every submitted written assignment is checked using a plagiarism software. We therefore suggest not to share solutions under any circumstances, as this may give rise to the suspicion of plagiarism.

1.1 Task 1: Customer Self-Service Portal - SwissAXA

Background: SwissAxa is an insurance company in Nord-Rhein Westfalen (NRW), as part of its digital transformation strategy, they have employed a Scrum Team to build a Customer Self-Service Portal at their touchpoint with their customers.

The Customer Self-Service Portal with the features below will enable their customers to interact with SwissAXA, such that the digitally savvy customers can gain positive experiences:

1. **myPolicies:** with drill down function to show
 - a. SwissAxa Policies: a list view of all policies that the client purchased from SwissAxa (Expiration date of each policy should be displayed) to enable SwissAXA to either extend the policy or recommend new product(s) to the customer (i.e. start a campaign).
 - b. External Policies: a list view of all policies the client bought from external insurance companies to enable SwissAXA suggest comparable product(s) to the customer using an AI-powered technology of your choice. Client should be able to upload the external policies.
2. **myDocument:** Client should be able to upload all documents required for policy and claims processing. SwissAxa. SwissAXA can be able to download those documents.
3. **myBank:** Client should be able to access their bank mobile app (Sparkasse, N26, Deutsche Bank, etc.) from the customer portal and initiate debit/credit transactions. Client can neither apply for loan via this channel nor obtain an overdraft.
4. **myServices:** with drill down function for:
 - a. Claims Management: Here the client should be able to file a claim, upload videos/photos of damages, capture geo location (integrated with google map) of damages.
 - b. Policy Management: Here the client should be able to access his/her policies, upgrade policies, and make policy change requests.
 - c. Contact Management: Customer can send Email to SwissAxa Customer Service Desk or to their Insurance Agent.

- d. Scheduling: Customer can book appointments with the SwissAXA Customer Service Desk or with their Insurance Agent.
5. **myInformation:** Should display client personal details, details like personal data - name, current address and correspondence address, email address, bank account number, phone number, etc. - should be editable.

1.2 Task 2: Campaign Management Software - Bayer-Westphalian

Background: As the Product Owner of a Business Intelligence unit at Bayer-Westphalian Insurance company, you are required to develop a campaign which enables customers to transfer their monetized life insurance pay-out to their grandchildren. This was meant to nudge the grandchildren to either underwrite their homeowner's insurance or buy into an investment fund. If the client's grandchild already has a homeowner's insurance or invested into a fund, other insurance products that Bayer-Westphalian offers could be sold to the grandchild.

While underwriting the life insurance, the customer would usually indicate his wish to transfer the pay-out to his grandchildren and provide their contact details. Sometimes, letters of consent are needed from the grandchildren or their guardians to communicate with the grandchildren when appropriate.

Please develop a Campaign Management Software which will also perform other forms of automated campaigns such as the following:

Customers:

- List of customers and prospects
- Search Customers (Note: apply an innovative search logic)
- Modify Customer properties (name, email, address)
- Remove / Delete Customers and prospects

Products:

- List of products offered by Bayer-Westphalian
- Search for Products (Note: apply an innovative search logic)
- Modify Product properties (Price, expiration, duration)
- Remove / Delete Products
- Product change request to the policy management

Schedules:

- Payment due date for making payments:
 - Send out monthly reminder at specified dates to customers
 - Green - First reminder
 - Yellow - Second reminder
 - Red - Third reminder (Customer will likely default this month's payment, send reminder)
- Product nearing expiration: choose a time delta:
 - 3 months: Start marketing 3 months before expiration

- 6 months: Start marketing 6 months before expiration
- 12 months: Start marketing 12 months before expiration

And follow-up with customers/policy holders who have opted to pay out to their beneficiaries with marketing any other product.

Campaigns:

- **Target Audience Segmentation:**
 - Segment your contact database based on various criteria such as demographics, location, behaviour, payment history, etc.
 - Create targeted campaigns for specific customer segments to improve engagement and conversion rates. For example, customers living in an area prone to hazards can receive critical warnings and alerts.
 - Keeping track who has been contacted to prevent duplicate or incessant marketing emails or props.
 - Following up with potential customers who have shown interest.
 - Remove uninterested parties (removal criteria must be set) from the mailing list.
- **Campaign Analytics:**
 - Track and measure the performance of your campaigns through key metrics such as engagement, conversions, ROI, etc.
 - Generate reports and visualizations to gain insights into campaign effectiveness.

1.3 Task 3: Learning Management System - NordWest Business School

Background: NordWest University strives to become the most digital private university in Germany. Part of its digital transformation strategy includes building a new Learning Management System (LMS) which can help increase positive user-experience at their lecturer/student touchpoints. At their Lecturer Lifetime Journey, the university identified four major stages which are:

- **meetMyStudent**
- **teachMyStudent**
- **testMyStudent**
- **gradeMyStudent**

Several digital technologies are introduced at those lifetime stages to make the learning experience more user-friendly and intuitive for both lecturers and students. One such technology is the lecturer's **dashboard**, which provides easy access to all the courses taught by the lecturer, as well as his calendar and weekly timetable. For those lecturers who may not be as familiar with digital tools, there are quick access buttons that provide access to almost every activity that can be performed on the LMS. Additionally, lecturers can have the ability to switch to the student portal whenever necessary.

At **meetMyStudent**, lecturers can manage their students in one place, organize them according to various criteria and assign them different tasks such as assignments, projects, and exams. The student overview shows students' names, course, semester level, their student ID. Lecturers can notify students when they are due to submit tasks

and can also send them reminders to register for both current and carry-over exams. Students can attend online classes, the lecturers can activate Teams to allow students to attend virtual classes. The student overview shows the name of the student(s), course, semester level, student ID, number of attempts, and whether a project has been assigned to the student. All scheduling is done with the help of a calendar.

In the second stage of the lifetime journey, lecturers are able to have a more digitalized teaching experience. This is the touchpoint where lecturers can upload all their teaching materials. For example, lecturers can upload their slides, course descriptions, question bank, and every other AI-powered material needed for positive learning experience. The uploaded materials can take different kinds of formats. The course description section displays the name of the course, an avatar, and a short description of the course. The slide-set section shows the title of the slide, the date of the lecture, and it can enable files like pdf, video, link, glossary, literature upload, etc.

In the third stage, **testMyStudent**, lecturers can create different classifications of examination questions, including multiple choice, essay, yes/no, and mathematical questions, among others. Each question carries a specific mark, and if the correct answers to the multiple choice questions or the yes/no questions are entered into the system before the exam, automatic marking is activated. For all other types of questions, an AI-powered technology can assess the student's response and assign a score based on predetermined weightings. Mathematical questions can be solved using any applicable AI-technology. Prior to releasing the exam questions to the students, lecturers have the option to edit and modify them. Additionally, lecturers can filter students who are eligible to sit for the exams based on course, semester, student ID, and number of attempts. To be able to sit for the examination, students must pass all coursework assigned for their Blended Learning.

The final stage of the lifecycle - **gradeMyStudent** - aims at giving lecturers access to the exams written by their students, mark them, display the results, and publish them. After the result is published, students are alerted by email and can launch into the student portal to view their results.

To-Dos

1. Identify your stakeholders, construct a stakeholder table, and prioritize them.
2. Construct your requirement engineering not only from stakeholders but also from supporting documents and existing systems (if given) using any requirement elicitation techniques. Document your findings in Confluence.
3. Distinguish your functional and non-functional specifications including your business rules.
4. From your analysis, come up with AI-powered features which you gathered from the stakeholders that will be added to the core features. Add them into your functional requirements.
5. Design your user-story mapping, define your product vision and show your product roadmap.
6. With the aid of any Unified Modelling Language (UML) or any other graphical method, communicate your To-Be Concept to the stakeholders.
7. Constitute a scrum team to manage the product development. Document your ceremonies in JIRA.
8. In your release plan, show your versioning and inform stakeholders your dates of release.
9. Develop your product using either any programming language of your choice or simple mock-ups.

2. ADDITIONAL INFORMATION FOR THE EVALUATION OF THE PROJECT REPORT

When conceptualizing and writing the project report, the evaluation criteria and explanations given in the writing guidelines should be considered.

3. TUTORIAL SUPPORT

In this project report task, several support channels are open; as the student, it is your responsibility to select your preferred support channel. The tutor is available for technical consultations and for formal and general questions regarding the procedure for processing the project report. However, the tutor is not required to approve outlines or parts of texts and drafts. Independent preparation is part of the examination work and is included in the overall evaluation. However, general editing tips and instructions are given in order to help you get started with the project report.

REFERENCES

- Fahsel, J.** (2023). *Techniques and Methods for Agile Software Development: IWNFO_1*. IU Internationale Hochschule GmbH.
- Bühne, S., & Herrmann, A.** (2022). *Handbook Requirements Management: Education and Training for IREB Certified Professional for Requirements Engineering Advanced Level Requirements Management*. IREB.
- Sommerville, I.** (2016). *Software Engineering* (10th ed.). Pearson.
- Brückmann, T.** (2022). *Requirements Engineering: IREN01*. IU Internationale Hochschule GmbH.